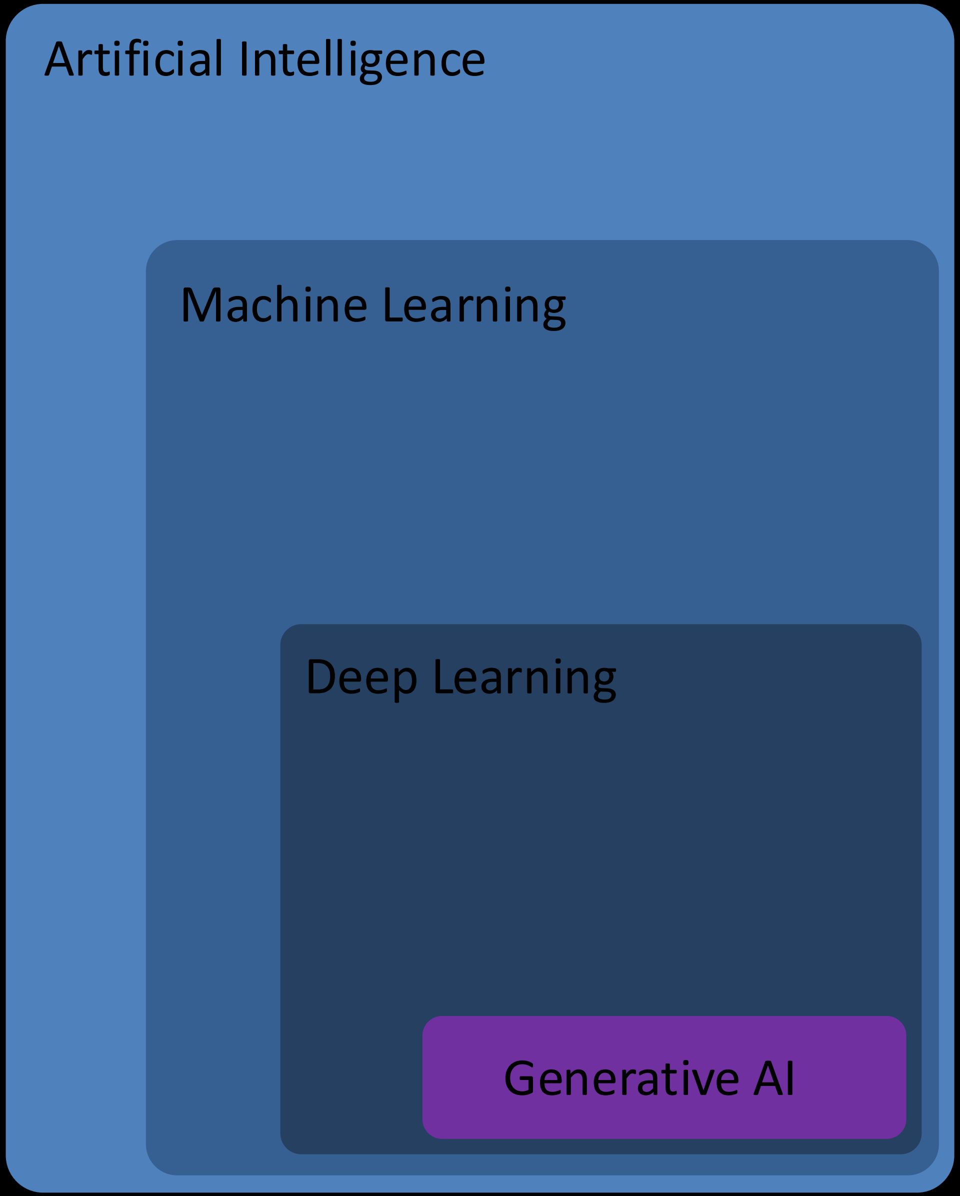
AI-Assisted Coding for Educators

AI-assisted coding

AI coding tools labs

Human thinking + AI collaboration

GenAI



1956

Artificial Intelligence

Intelligence demonstrated by machines

1997

Machine Learning

Learn from data

2017

Deep Learning

Model after the human brain (Neural Networks)

2021

Generative AI

Create new written, visual, and auditory content

Generative AI

The best thing about AI is its ability to …

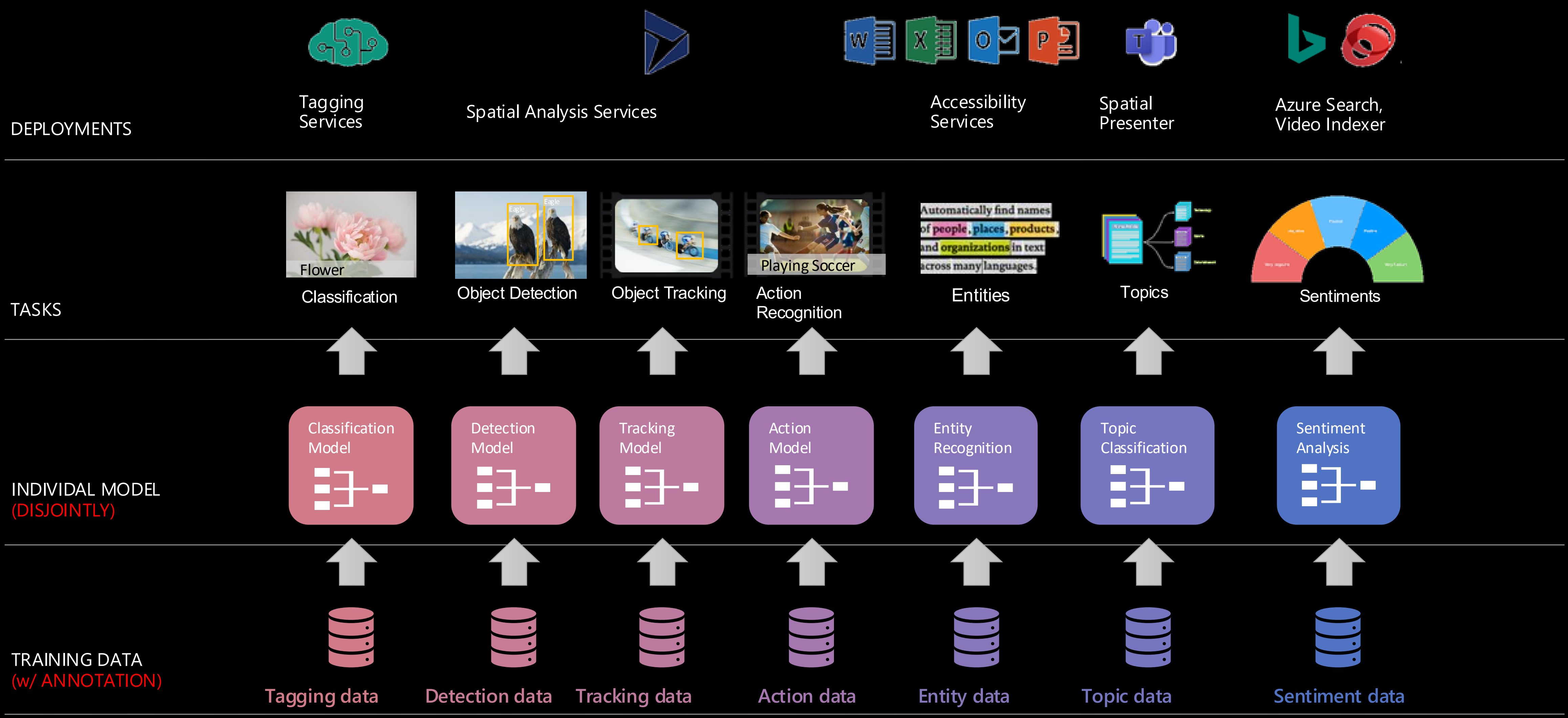
Adapt = 30%

Process = 22%

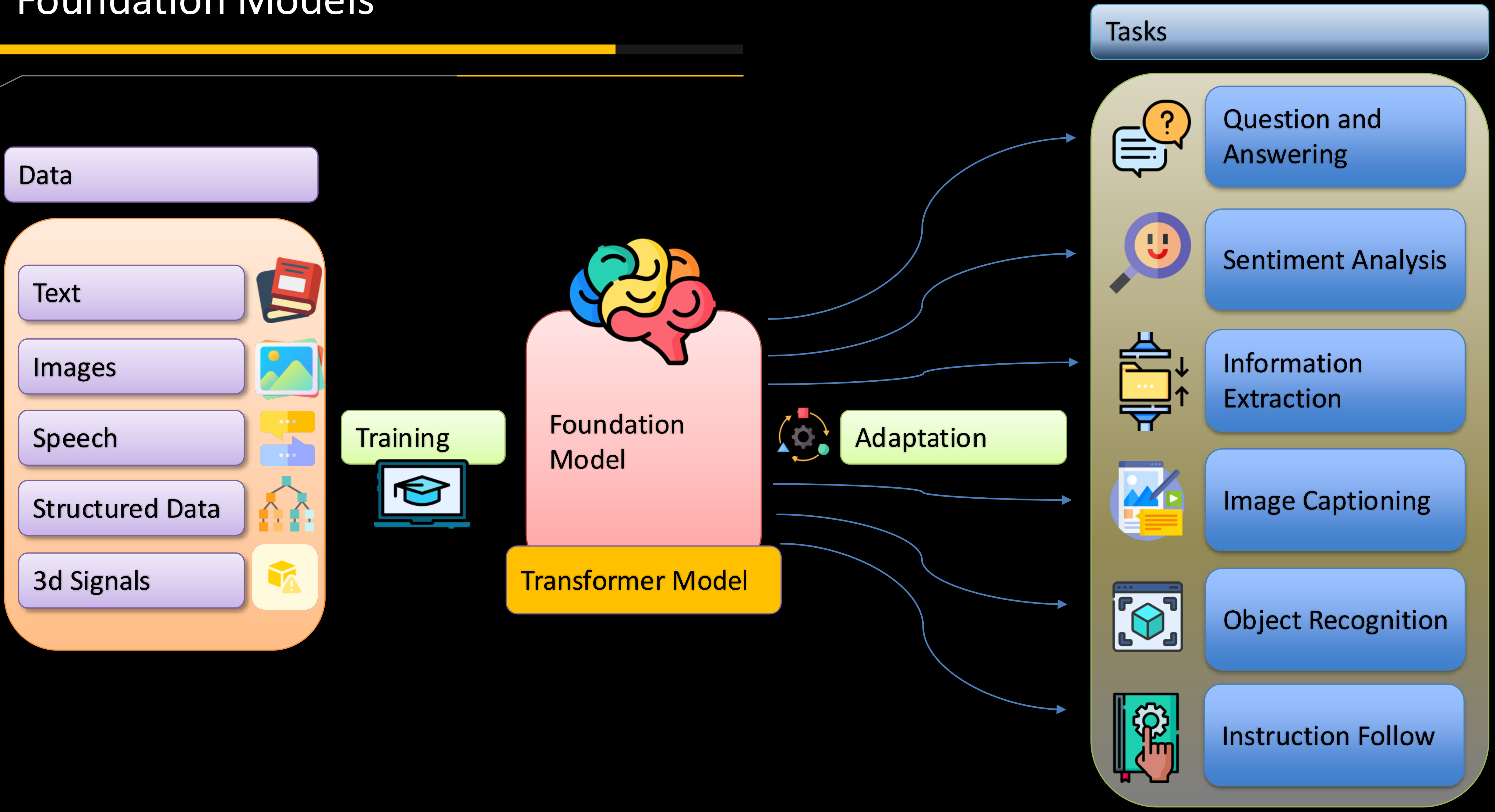
Analyze = 7%

Traditional model development

High cost & slow deployment - Each service is trained disjointly



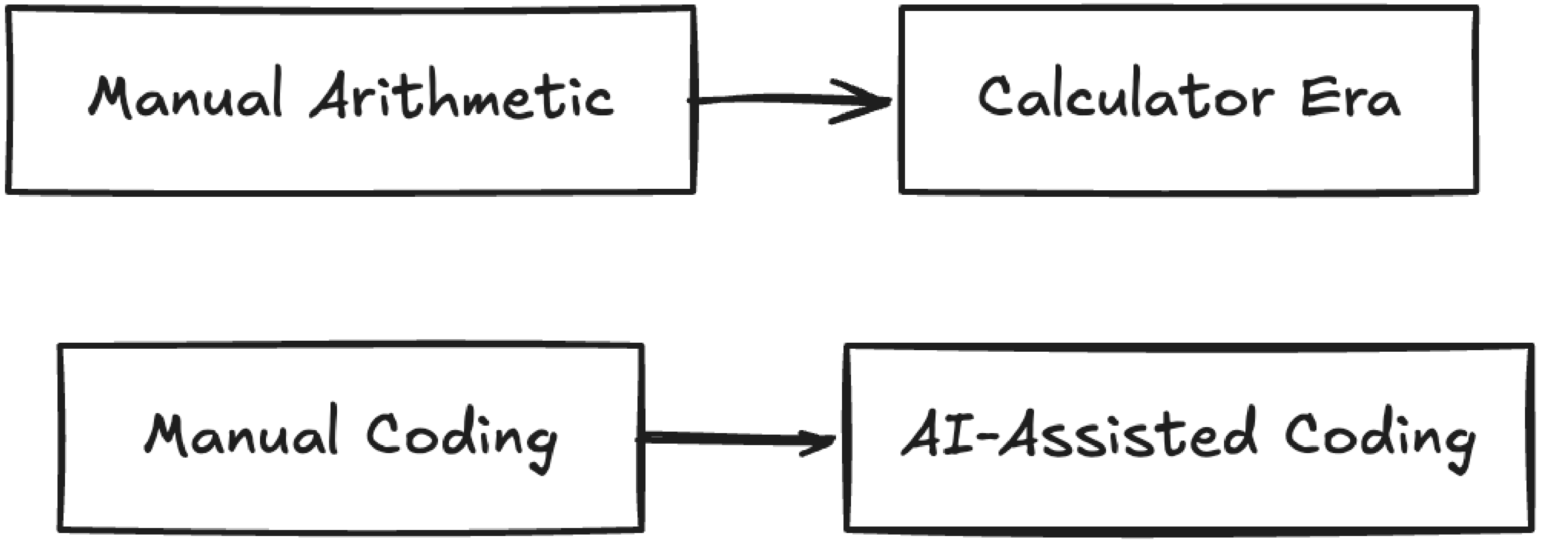
Foundation Models



Why This Matters

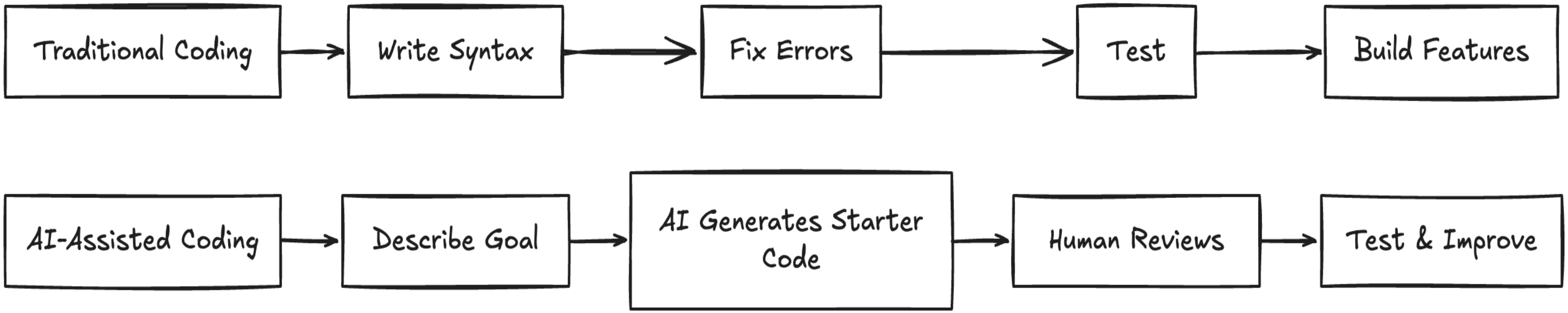
- AI is changing software development
- Students will work with AI professionally
- Goal: teach thinking, not memorization

Calculator Analogy



- Calculators changed math education
- AI coding tools change (programming) education
- Humans still provide reasoning and verification

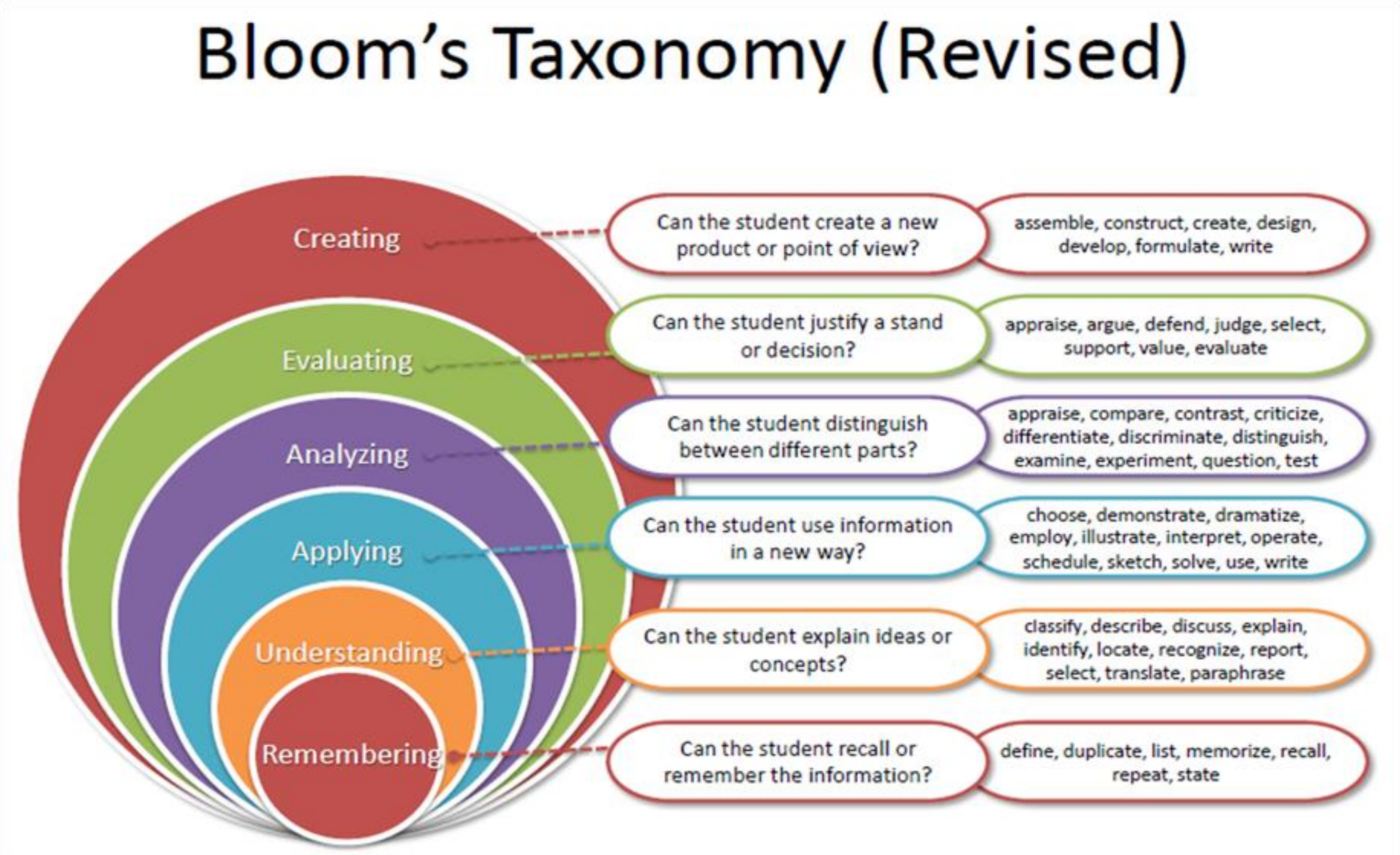
Traditional vs AI-Assisted Coding



- Before AI: syntax, debugging, boilerplate
- With AI: design, testing, evaluation
- Shift toward higher-order thinking

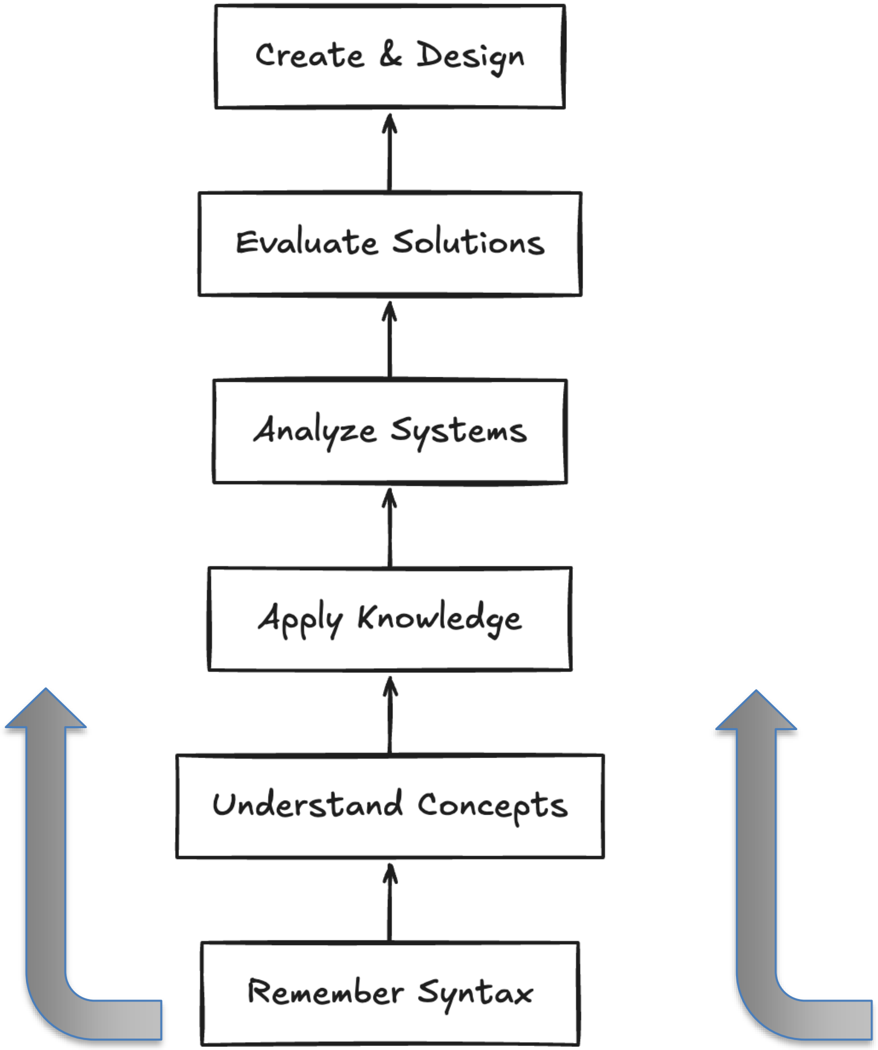
Design

- Learning objectives
- Select instructional strategies
- Select assessment methods



Bloom's Taxonomy Shift

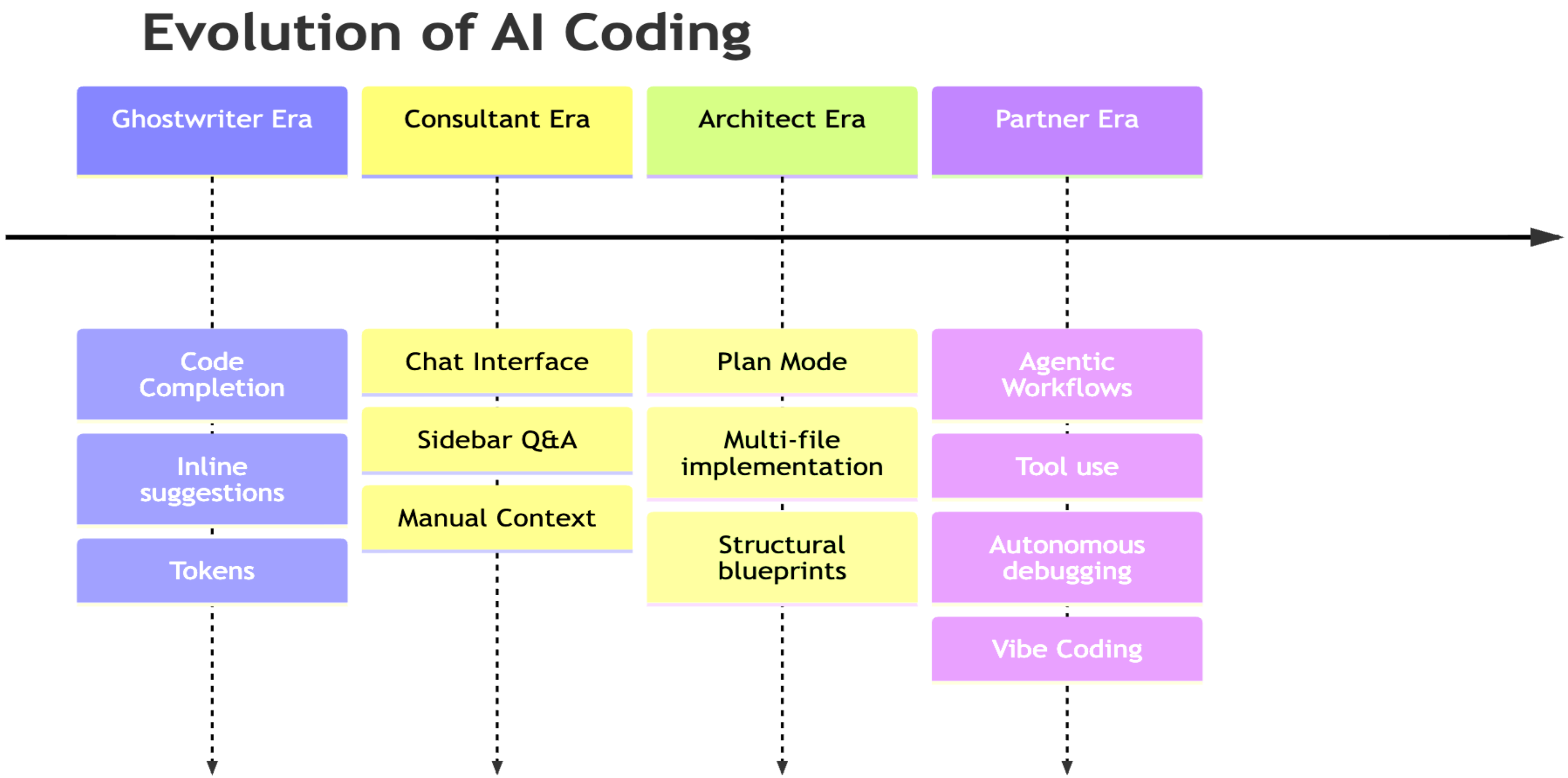
- Less time on syntax
- More time on analysis and design
- AI can elevate classroom thinking



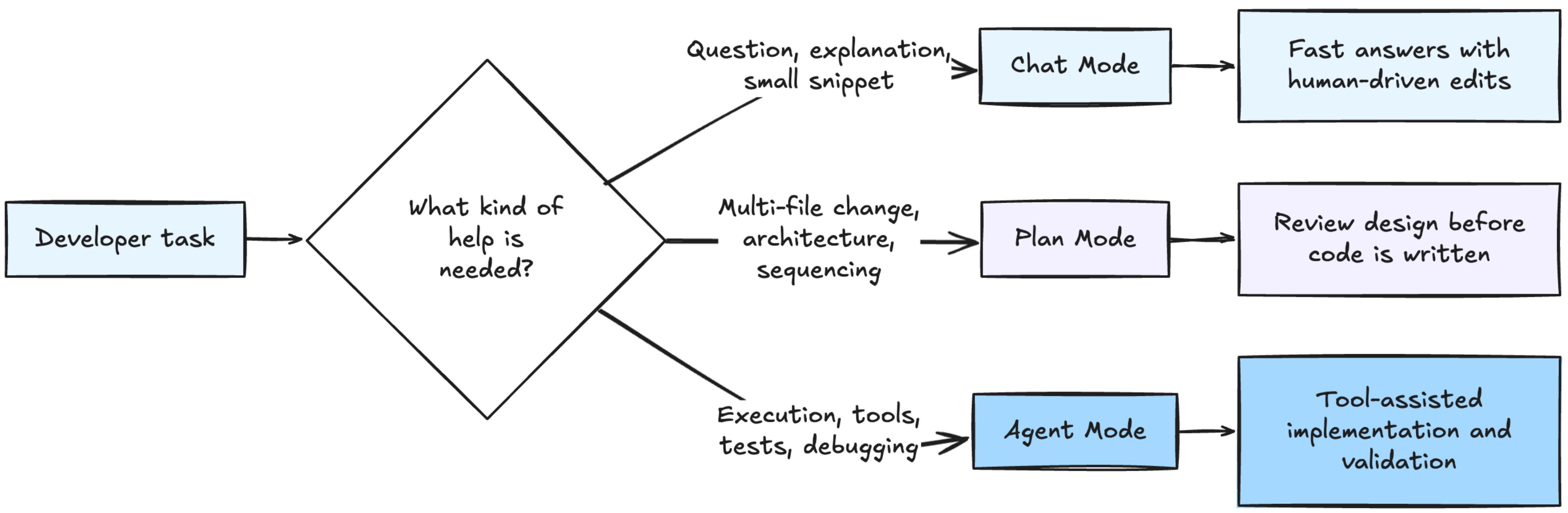
SAMR Model (Enhancement-Transformation)

- Substitution
- Augmentation
- Modification
- Redefinition

Tooling transformation



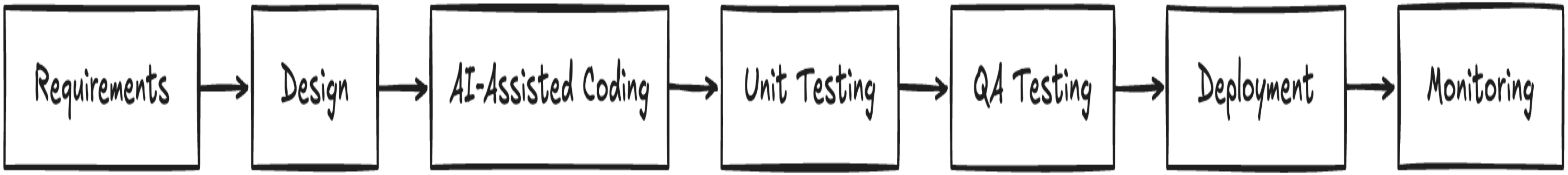
IDE Modes for AI-Assisted Coding



Lab -1

- VS code + Continue + Ollama
  - Chat
  - Plan
  - Agent

Fibonacci Example

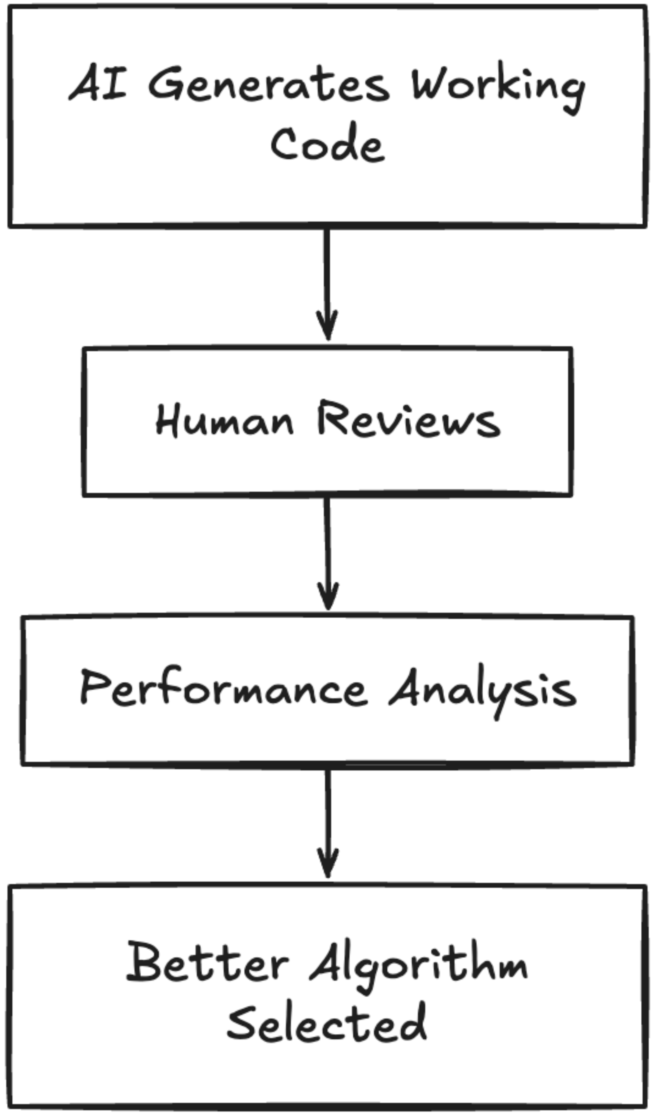


- Tiny example, full SDLC
- Requirements → Design → Testing
- AI helps implementation

Lab 2

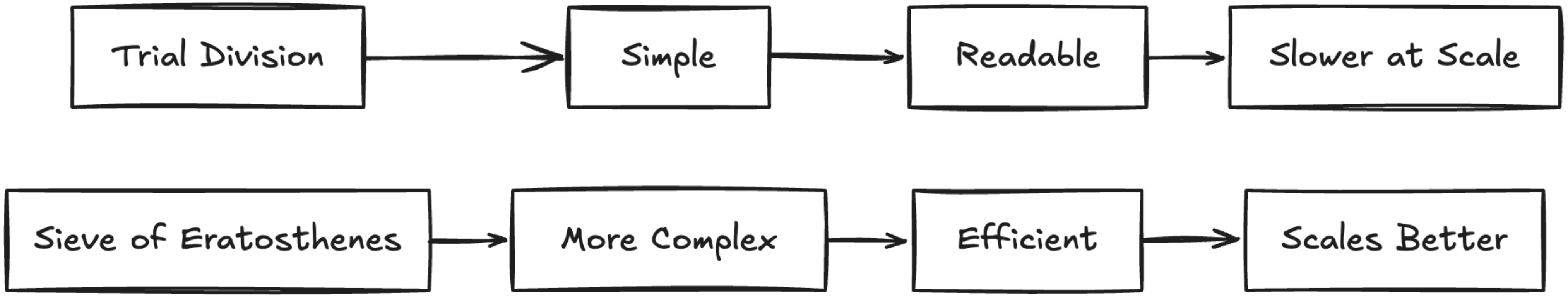
Prime Numbers Example

- AI often suggests simpler solutions first
- Humans improve scalability
- Correct ≠ optimal



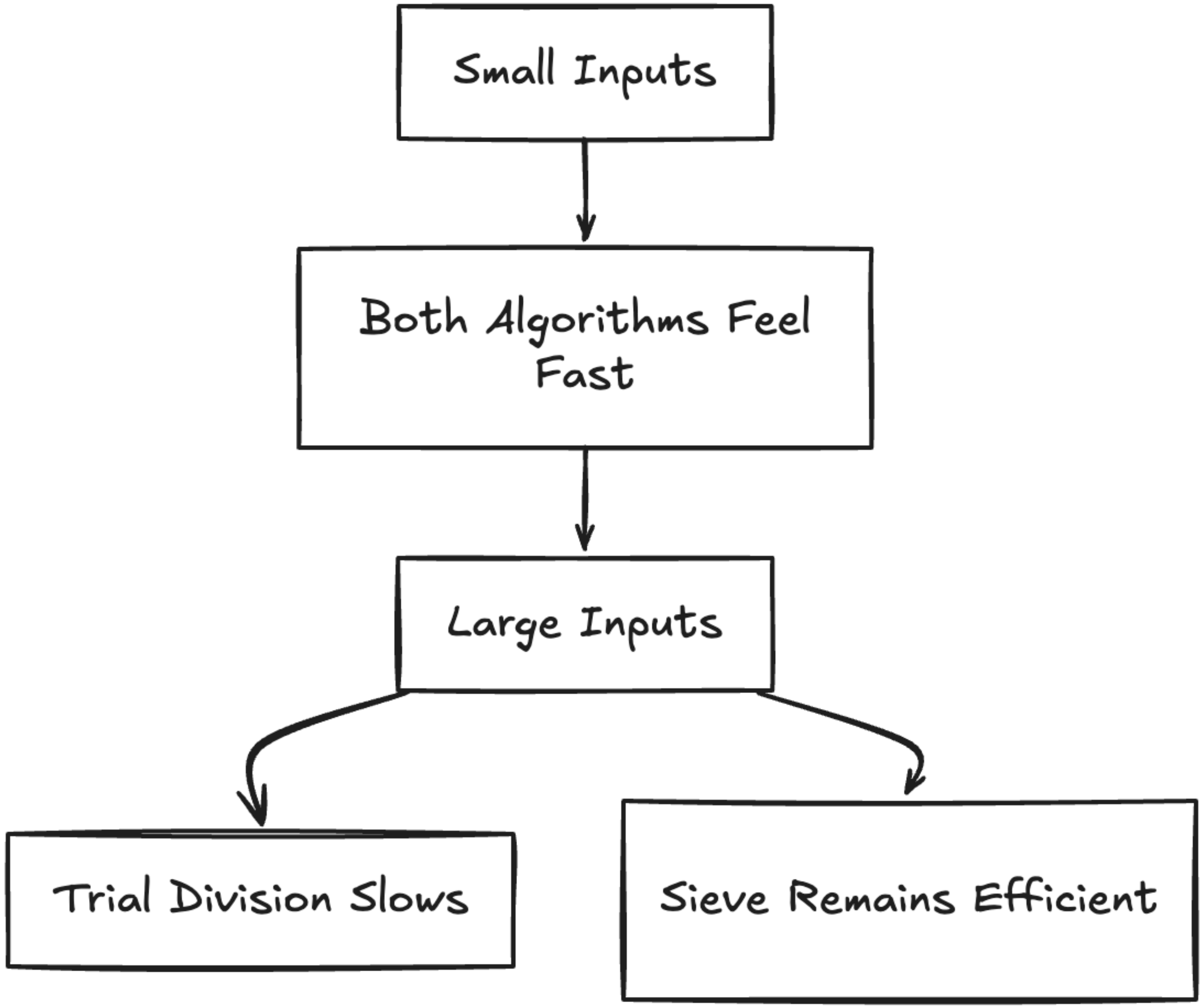
Trial Division

- Readable and beginner friendly
- Time complexity: $O(n\sqrt{n})$
- Slows dramatically for large inputs



Sieve of Eratosthenes

- More advanced algorithm
- Time complexity: O(n log log n)
- Much faster for large inputs



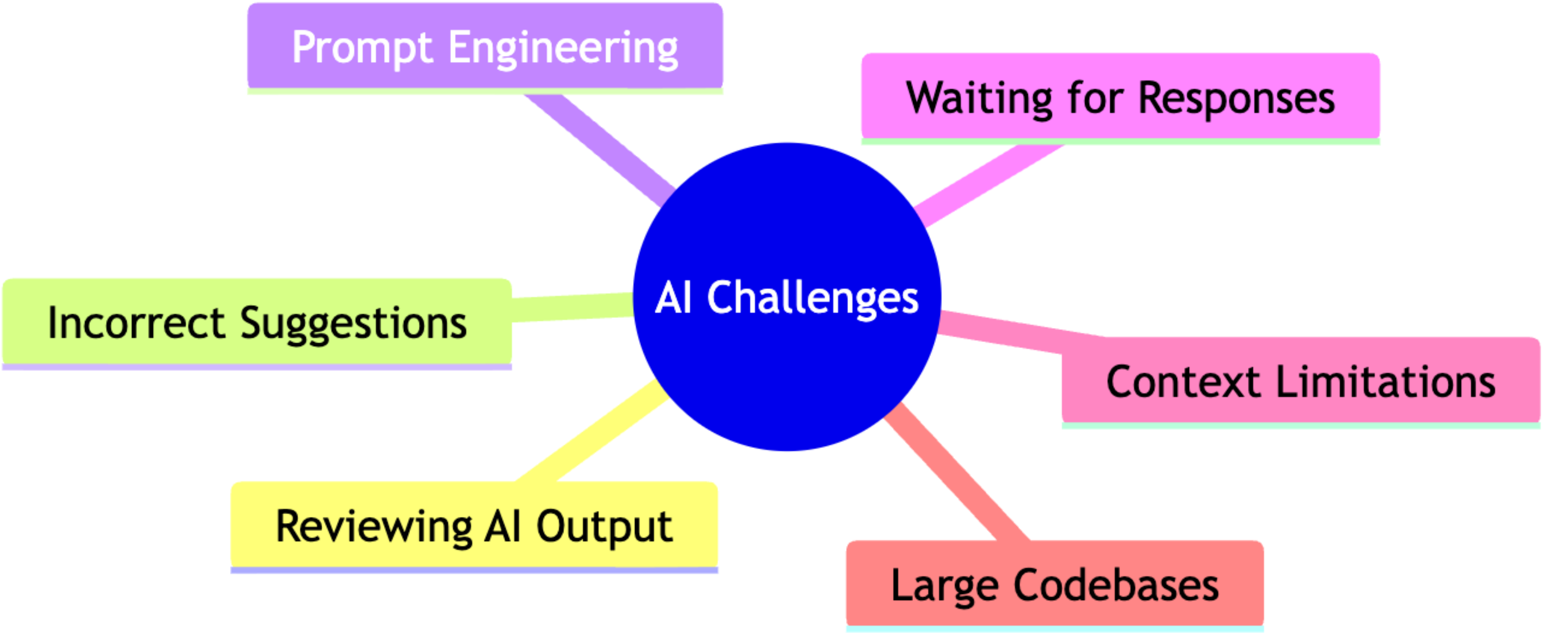
Risks

Research Findings (METR Study)

- Experienced developers used AI
- Tasks sometimes took 19% longer
- AI helps, but oversight matters

https://metr.org/blog/2025-07-10-early-2025-ai-experienced-os-dev-study/

Research Findings (METR Study)



https://metr.org/blog/2025-07-10-early-2025-ai-experienced-os-dev-study/

Risk mitigation plan for AI-assisted Coding in Education

Key Concerns

Blind trust

Prevent shortcut learning

Demystify the tool

Safeguards

Verify AI code & understand it before use

Use manual coding & AI to build foundational skills

Teach how AI coding tools work

Risk mitigation plan for AI-assisted Coding in Education

Key Concerns

Skills gap

No human-in-the loop

Safeguards

Foster critical thinking & peer reviews to reduce

overconfidence in unverified outputs

Mandate code reviews & discussions on AI-generated code

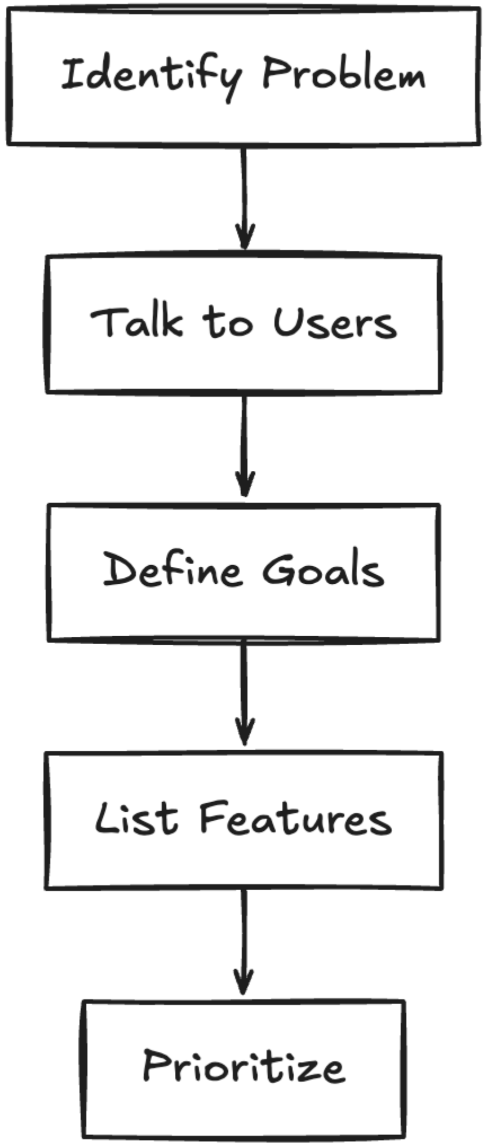
Lab 3

- Safe vibe coding

Best practices

- Supporting resources:
- Prompt Engineering Examples
- Design Note Examples

Design Thinking



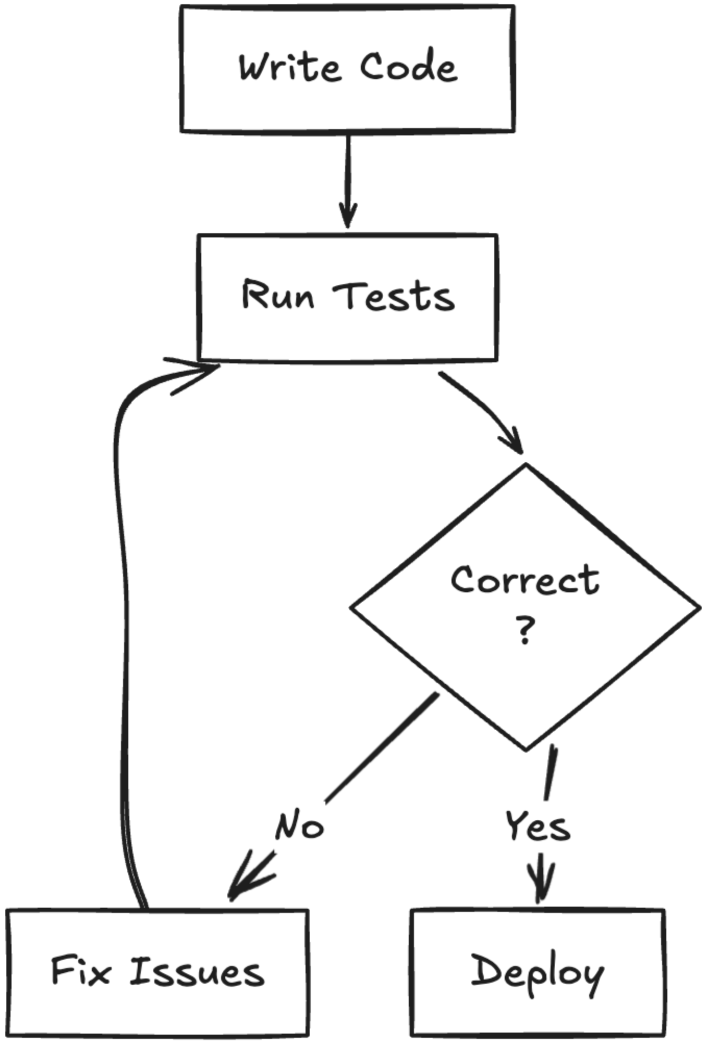
- Plan before coding
- Compare possible solutions
- Think about scalability and usability

Development with AI

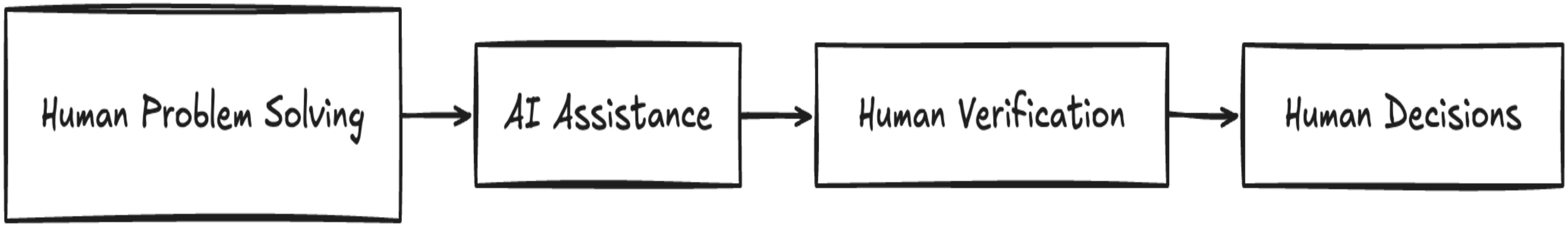
- AI generates starter code
- Students review and modify
- Human judgment remains essential

Testing Matters

- Unit testing checks functions
- QA testing checks user experience
- AI can generate tests, humans verify

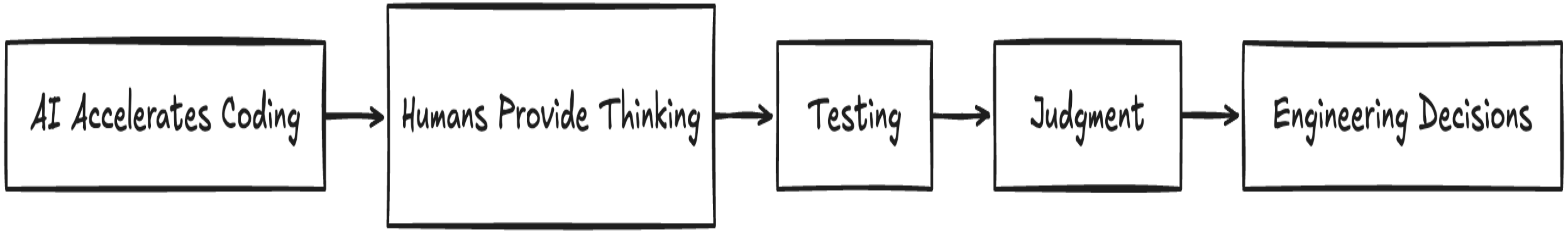


Human Thinking vs AI Suggestions



- AI accelerates implementation
- Humans optimize and evaluate
- Engineering requires tradeoffs

Key Message for Educators

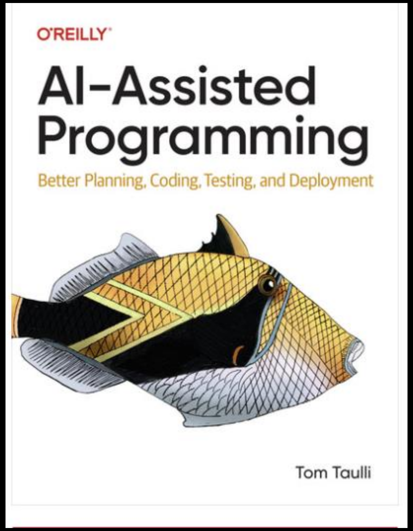


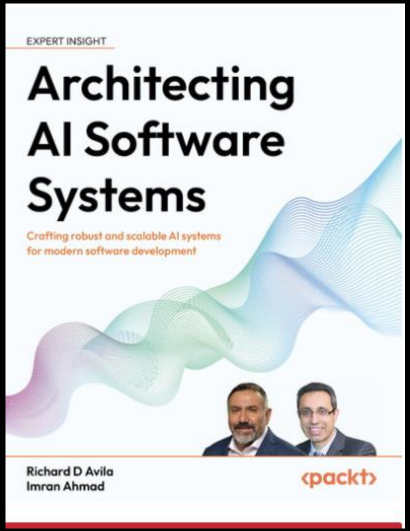
- AI does not replace thinking
- AI reduces mechanical friction
- Teach students to think critically with AI

Discussion Questions

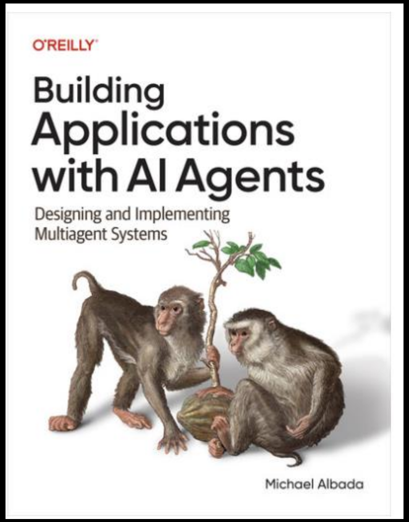
- What should AI automate?
- What thinking should humans always do?
- How do we teach responsible AI use?

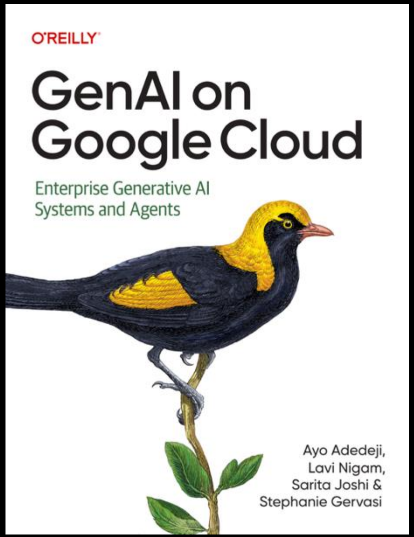
Resources

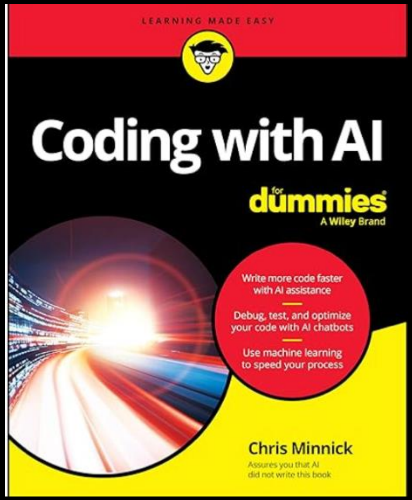

O'REILLY
AI-Assisted Programming
Better Planning, Coding, Testing, and Deployment
Tom Taulli


EXPERT INSIGHT
Architecting AI Software Systems
Crafting robust and scalable AI systems for modern software development
Richard D Avila
Imran Ahmad
<packt>


O'REILLY
Generative AI for Software Development
Building Software Faster and More Effectively
Sergio Pereira


O'REILLY
Building Applications with AI Agents
Designing and Implementing Multiagent Systems
Michael Albada


O'REILLY
GenAI on Google Cloud
Enterprise Generative AI Systems and Agents
Ayo Adedeji, Lavi Nigam, Sarita Joshi & Stephanie Gervasi


LEARNING MADE EASY
Coding with AI
for dummies
A Wiley Brand
Write more code faster with AI assistance
Debug, test, and optimize your code with AI chatbots
Use machine learning to speed your process
Chris Minnick
Assures you that AI did not write this book